

Name:- Jaykishan J Saharan

Div:- D Roll no:- 26

course/program:- BTech, CSE SY

Subject :- Discrete Maths

DM assignment. 3.

Q.1) Define algebraic structure, binary operations, and closure property with examples.

Solⁿ:-

i. Algebraic Structure:-

An algebraic structure consists of a non-empty set and one or more binary operations defined on that set, which must follow certain rules (like closure).

example:- The set of integers ($\mathbb{Z}$) combined with the operation of addition (+) is an algebraic structure, denoted as $(\mathbb{Z}, +)$

- Set: The set is the integers. ($\mathbb{Z} = \{-1, 0, 1, 2\}$)
- Binary operation: The operation is addition (+).
- closure: For any two integers a and b , their sum $a+b$ is always another integer. This is the closure property, which this structure satisfies.

ii Binary Operations:

A binary operation, denoted by $*$, on a non-empty set A is a function from $A \times A \rightarrow A$. This means that for any two elements a and b in A , the result of the operation, $a * b$, is a unique element that also belongs to A .

eg: Addition on the set of integers ($\mathbb{Z}$):

when we add any two integers, the result is always another integer. for example, $5 + (-3) = 2$ and 2 is integer.

iii. Closure property:

The Closure property is a fundamental condition for a binary operation, stating that the operation on any two elements of a set always produces an element that is also in that same set. This property is a prerequisite for defining more complex algebraic structures.

example of closure:

- Addition on natural numbers ($\mathbb{N}$): The set of natural numbers $\{1, 2, 3, 4\}$ is closed under condition, for any two natural numbers a and b , their sum $a + b$

Q.2) Show that $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$ is group under addition modulo 5.

Solⁿ:-

$+_5$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	5
2	2	3	4	5	1
3	3	4	5	1	2
4	4	5	1	2	3

i. closure property.

$$\forall a, b \in \mathbb{Z}_5$$

$$a +_5 b \in \mathbb{Z}_5$$

$\therefore$ satisfies closure property.

ii Associative property.

L.H.S $(a +_5 b) +_5 c = a +_5 (b +_5 c)$ R.H.S

ex:

$$(0 +_5 1) +_5 2 = 1 +_5 2 = 3$$

$$0 +_5 (1 +_5 2) = 0 +_5 3 = 3$$

$$\therefore (a +_5 b) +_5 c = a +_5 (b +_5 c)$$

$\therefore$ Satisfies associative property.

iii Identity Property.

$\rightarrow a +_5 e = a$

eg:

$$1 +_5 e = 1$$

$$e = 1 - 1$$

$$\boxed{e=0} \in \mathbb{Z}$$

$\therefore$ Satisfies Identity property.

iv. Inverse property.

$\rightarrow a a^{-1} = e$

$$1 a^{-1} = 0$$

$$a^{-1} = 0 \in \mathbb{Z}$$

$\therefore$ Satisfies Inverse property.

$\therefore$ It satisfies all required properties: closure, associative, identity, Inverse.

$\therefore (\mathbb{Z}, +_5)$ is a group.

Q.3) Consider the encoding function $E: B^3 \rightarrow B^8$ defined by
 $E(000) = 00000000$, $E(001) = 10111000$, $E(010) = 00101101$
 $E(011) = 10010101$, $E(100) = 10100100$, $E(101) = 10001001$
 $E(110) = 00011100$, $E(111) = 00110001$.

Find the minimum hamming distance between the code words. Find the no. of errors which can be detected and corrected.

Solⁿ:-

Hamming distance:

- $d(000, 001) = d(00000000, 10111000) = 4$
- $d(000, 010) = d(00000000, 00101101) = 4$
- $d(000, 011) = d(00000000, 10010101) = 4$
- $d(000, 100) = d(00000000, 10100100) = 3$
- $d(000, 101) = d(00000000, 10001001) = 3$
- $d(000, 110) = d(00000000, 00011100) = 3$
- $d(000, 111) = d(00000000, 00110001) = 4$
- $d(001, 010) = d(10111000, 00101101) = 5$
- $d(001, 011) = d(10111000, 10010101) = 4$
- $d(001, 100) = d(10111000, 10100100) = 2$
- $d(001, 101) = d(10111000, 10001001) = 4$
- $d(001, 110) = d(10111000, 00011100) = 4$
- $d(001, 111) = d(10111000, 00110001) = 3$
- $d(010, 011) = d(00101101, 10010101) = 4$

Similarly we will all possible pair distance
 The shortest comes is 2.

$d = 2$.

→ Determine error

A code with a minimum Hamming distance of d_{min} can detect up to S errors, where

$$S = d - 1$$

• Number of detectable errors (S) = $d - 1 \Rightarrow 2 - 1 \Rightarrow 1$.

The code can detect up to 1 error.

→ Determine error correction,

$$t = \frac{d-1}{2}$$

$$\text{No. of correctable error} = \frac{2-1}{2} = \frac{1}{2} \approx 0$$

∴ The code can correct 0 errors.

Minimum Hamming distance = 2

Number of detectable error = 1

Number of correctable error = 0

9.4) Show that the $(3,6)$ encoding function $e: B^3 \rightarrow B^6$ defined by
 $e(000) = 000000$, $e(001) = 001100$, $e(010) = 010011$
 $e(011) = 011111$, $e(100) = 100101$, $e(101) = 101001$
 $e(110) = 110110$, $e(111) = 111010$ is group code.

Solⁿ:-

i. closure property.

$$\forall a, b \in e$$

eg:-

$$e(000) + e(001) = e(011) \in e$$

∴ satisfies closure property.

ii. Associative property.

$$\rightarrow (e(001) + e(010)) + e(011) = e(001) + (e(010) + e(011))$$

$$(e(001) + e(010)) + e(011) = (001100 + 010011) + 011111$$
$$= 011111 + 011111$$

$$= 000000$$

$$e(001) + (e(010) + e(011)) = 001100 + (010011 + 011111)$$

$$= 001100 + 001100$$

$$= 000000$$

$$L.H.S = R.H.S$$

$\therefore$ Satisfies associative property.

iii. Identity property.

$$e(001) + E = e(001)$$

$$001100 + E = 001100$$

$$E = 0 = e(000) \in e$$

$\therefore$ Satisfies Identity property.

iv. Inverse property.

$$e(001) \cdot e(001)^{-1} = E$$

$$001100 \cdot e(001)^{-1} = 0$$

$$e(001)^{-1} = 0 = e(000) \in e$$

$\therefore$ Satisfies Inverse property.

$(e, *)$ is a group.

Q.5) Find Hamming Weight of the given words.

i. 1010101 ii. 1110011 iii. 1001101

Solⁿ:- Hamming weight, $d =$

$$d(1010101, 1110011) = 1010101 + 1110011 = 0100110 = 3$$

$$d(1010101, 1001101) = 1010101 + 1001101 = 0011000 = 2$$

$$d(1110011, 1001101) = 1110011 + 1001101 = 0111110 = 5$$

$\therefore d(1010101, 1001101)$ is the hamming weight

$$\text{Smallest} = \underline{\underline{2}}$$

Q.6) Show that the group (a) $\{(1,2,3,4)^* 5\}$ and

(b) $\{(1,2,3,4,5,6)^* 7\}$ is cyclic. Note (* = multiply).

Solⁿ:- (a) $\{(1,2,3,4)^* 5\}$

x_5	1	2	3	4
1	1	2	3	4
2	2	4	1	3
3	3	1	4	2
4	4	3	2	1

$$(b) = \{(1,2,3,4,5,6)^* 7\}$$

3) $3^1 = 3$

$$3^2 = 3 \times_7 3 = 2$$

$$3^3 = 3^2 \times_7 3 = 2 \times_7 3 = 6$$

$$3^4 = 3^3 \times_7 3 = 6 \times_7 3 = 4$$

$$3^5 = 3^4 \times_7 3 = 4 \times_7 3 = 5$$

$$3^6 = 3^5 \times_7 3 = 5 \times_7 3 = 1$$

Thus (b) is generated by element 3 of set (b).

$\therefore$ It is a cyclic group.

$$(a) = \{ (1, 2, 3, 4) * 5 \}$$

$$= 3^1 = 3$$

$$3^2 = 3 \times_5 3 = 4$$

$$3^3 = 3^2 \times_5 3 = 4 \times_5 3 = 2$$

$$3^4 = 3^3 \times_5 3 = 2 \times_5 3 = 1$$

$\therefore (a)$ is generated by element 3

$\therefore$ It is a cyclic group.

Q.7) Show that the set $G = \{1, 2, 3, 4, 5, 6\}$ forms an Abelian group with respect to multiplication modulo 7. Is it cyclic?

Solⁿ:-

$\times_7$	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	1	3	5
3	3	6	2	5	1	4
4	4	1	5	2	6	3
5	5	3	1	6	4	2
6	6	5	4	3	2	1

i. Closure property.

$$\forall a, b \in G$$

$$\text{eg: } 1 \times_7 4 = 4 \in G$$

$\therefore$ Satisfies closure property.

ii Associative property.

$$(a \times_7 b) \times_7 c = a \times_7 (b \times_7 c)$$

eg. $(1 \times_7 2) \times_7 3 = 1 \times_7 (2 \times_7 3)$

$$(1 \times_7 2) \times_7 3 = 2 \times_7 3 = 6$$

$$1 \times_7 (2 \times_7 3) = 1 \times_7 6 = 6$$

$\therefore$ Satisfies associative property.

iii Identity property.

$$a \times_7 e = a$$

$$2 \times_7 e = 2$$

$$\boxed{e=1} \in G$$

$\therefore$ Satisfies Identity property.

iv. Inverse property.

$$a a^{-1} = e$$

$$2 a^{-1} = 1$$

$$a^{-1} = \frac{1}{2} \approx 0 \in G$$

$\therefore$ Satisfies Inverse property.

v. Commutative property.

$$a \times_7 b = b \times_7 a$$

$$1 \times_7 2 = 2 \times_7 1 = 2$$

$\therefore$ Satisfies commutative property.

$\therefore (G, \times_7)$ is an abelian group.

Q.8) find the code words generated by parity check matrix

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \text{ when encoding function is } E: B^3 \rightarrow B^5$$

Solⁿ:-

$$m=3, n=5$$

$$E: B^3 \rightarrow B^5$$

$$E: B^m \rightarrow B^n$$

$$\text{Rewrite, } H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Generator matrix, G

$$G = [I_m | A] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

$$B^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$$

$$e(w) = wG$$

$$e(000) = [000] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [0 \ 0 \ 0 \ 0 \ 0]$$

$$e(001) = [001] \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix} = [0 \ 0 \ 1 \ 0 \ 1]$$

$$e(010) = [010] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [0 \ 1 \ 0 \ 1 \ 1]$$

$$e(011) = [011] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [0 \ 1 \ 1 \ 1 \ 0]$$

$$e(100) = [100] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [1 \ 0 \ 0 \ 1 \ 0]$$

$$e(101) = [101] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [1 \ 0 \ 1 \ 0 \ 1]$$

$$e(110) = [110] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [1 \ 1 \ 0 \ 0 \ 1]$$

$$e(111) = [111] \begin{bmatrix} 1 & 1 & - \\ - & 1 & - \\ - & - & - \end{bmatrix} = [1 \ 1 \ 1 \ 0 \ 0]$$

The codewords generated are 00000, 00101, 01011,
01110, 10010, ~~10011~~, 11001, 11100
0111

8.9) $H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H .

(a) 10100 (b) 01101 (c) 11011

Sol:-

Parity check matrix, $H = [A^T | I_3]$

$$H = \left[\begin{array}{ccc|ccc} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

Received word $\Rightarrow H \cdot (r)^T$

(a) $H \cdot (r)^T = r^T = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$$H \cdot (r)^T = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

(b) 01101 = r

$$r^T = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

$$H \cdot (r)^T = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(c) r = 11011

$$r^T = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$H \cdot (r)^T = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Q.10) Consider the $(3,6)$ group encoding function $e: B^3 \rightarrow B^6$ defined by $e(000) = 000000$, $e(001) = 000110$, $e(010) = 010010$, $e(011) = 010100$, $e(100) = 1001001$, $e(101) = 100011$, $e(110) = 110111$, $e(111) = 110001$

Decode the following words relative to a maximum likelihood decoding function.

(a) ~~Solⁿ~~: 011110 (b) 101011 (c) 110010

Solⁿ:

(a) 011110

$$e(011) = 010100$$

In 3rd and 5th position an error occurred

$\therefore$ 2 error

The transmitted word is 1010100

The original message is 010

(b) 101011

$$e(101) = 100011$$

In 3rd position an error occurred

$\therefore$ 1 error

The transmitted word is 100011

The original message is 100

(c) 110010

$$e(110) = 110111 \quad \& \quad e(111) = 110001$$

In 4th & 6th position

In 5th and 6th position errors

$\therefore$ 2 error

The transmitted word is 110111 or 110001

The original message is 110